U1
Waveshare ESP32-S3-Nano-M 16MB flash 8MB PSRAM
THREAD_BOOT JP1.1
JP1.2
JP1.3
I2C_SDA JP1.4
I2C_SCL JP1.5
JP1.6
RF_GDO0 JP1.7
SPI_MOSI JP1.8
SPI_MISO JP1.9
SPI_SCK JP1.10
ETH_CS JP1.11
JP1.12
S3_BOOT_SERVICE JP1.13
GND JP1.14
+5V_S3 JP1.15
GPIO48
3V3
GPIO46
GPIO1
GPIO2
GPIO3
GPIO4
GPIO11
GPIO12
GPIO13
GPIO14
VBUS
GPIO0
GND
VIN
GPIO43 JP2.1 THREAD_RX
GPIO44 JP2.2 THREAD_TX
EN JP2.3
GND JP2.4 GND
GPIO5 JP2.5 RF_GDO2
GPIO6 JP2.6 RF_CS
GPIO7 JP2.7 ZB_CTS
GPIO8 JP2.8 RADAR_RX
GPIO9 JP2.9 ETH_RESET
GPIO10 JP2.10 ETH_INT
GPIO17 JP2.11 ZB_RX
GPIO18 JP2.12 ZB_TX
GPIO21 JP2.13 ZB_RTS
GPIO38 JP2.14 RADAR_TX
GPIO47 JP2.15 THREAD_EN

U2
ESP32-H2-DevKitM-1-N4S
THREAD_BOOT J1.1
J1.2
J1.3
J1.4
J1.5
J1.6
J1.7
J1.8
J1.9
J1.10
J1.11
J1.12
GND J1.13
+5V_H2 J1.14
GND J1.15
3V3
EN
GPIO0
GPIO1
GPIO2
GPIO3
GPIO13
GPIO14
GPIO4
GPIO5
NC
VBAT
GND
5V
GND
GND J3.1
GPIO24 J3.2 THREAD_TX
GPIO23 J3.3 THREAD_RX
GPIO10 J3.4
GPIO11 J3.5
GPIO25 J3.6
GPIO12 J3.7
GPIO8 J3.8
GPIO22 J3.9
GND J3.10 GND
GPIO9 J3.11 THREAD_BOOT
GND J3.12 GND
GPIO27 J3.13
GPIO26 J3.14
GND J3.15 GND

U3
Seeed XIAO MG24 pre-soldered 102010690
ZB_BOOT_SERVICE
NC
NC
NC
ZB_CTS 5
ZB_RTS 6
ZB_TX 7
ZB_BOOT_SERVICE
NC
NC
NC
ZB_CTS
ZB_RTS
ZB_TX
ZB_RX 8 ZB_RX
NC 9
NC 10
NC 11
NC 12
GND 13 GND
+5V_ZB 14 +5V_ZB

U4
WIZ850io finished module
GND J1.1
GND J1.2
SPI_MOSI J1.3
SPI_SCK J1.4
ETH_CS J1.5
ETH_INT J1.6
GND
GND
SPI_MOSI
SPI_SCK
ETH_CS
ETH_INT
GND J2.1 GND
+3V3 J2.2 +3V3
+3V3 J2.3 +3V3
NC J2.4
ETH_RESET J2.5 ETH_RESET
SPI_MISO J2.6 SPI_MISO

J1
FUSED ISOLATED 5V INPUT
+5V 1
GND 2
+5V
GND

J2
D24V22F3 external assembled buck
+5V 1
GND 2
+3V3 3
GND 4
+5V
GND
+3V3
GND

J3
DFR0627 WK2132 ready module harness
+3V3 1
GND 2
I2C_SDA 3
I2C_SCL 4
ZWAVE_RX 5
ZWAVE_TX 6
+3V3
GND
I2C_SDA
I2C_SCL
ZWAVE_RX
ZWAVE_TX

J4
ZAC93 ready module harness
+3V3 1
GND 2
GND 3
ZWAVE_RX 4
ZWAVE_TX 5
+3V3
GND
GND
ZWAVE_RX
ZWAVE_TX

J5
CC1101 ready module harness
+3V3 1
GND 2
SPI_SCK 3
SPI_MOSI 4
SPI_MISO 5
RF_CS 6
RF_GDO0 7
RF_GDO2 8
+3V3
GND
SPI_SCK
SPI_MOSI
SPI_MISO
RF_CS
RF_GDO0
RF_GDO2

J6
LD2450 radar harness
+5V 1
GND 2
RADAR_RX 3
RADAR_TX 4
+5V
GND
RADAR_RX
RADAR_TX

J7
BME680 external pod harness
GND 1
+3V3 2
I2C_SDA 3
I2C_SCL 4
GND
+3V3
I2C_SDA
I2C_SCL

D1
1N5819 USB supply isolation
+5V_S3 1
2 +5V

D2
1N5819 USB supply isolation
+5V_H2 1
2 +5V

D3
1N5819 USB supply isolation
+5V_ZB 1
2 +5V

R1
4.7k RF GDO0 pull-up
+3V3 1
2 RF_GDO0

R2
10k RF CS idle pull-up
+3V3 1
2 RF_CS

J8
XIAO BOOT unpopulated service pads
ZB_BOOT_SERVICE
GND 2
ZB_BOOT_SERVICE
GND

J9
Nano BOOT unpopulated service pads
S3_BOOT_SERVICE
GND 2
S3_BOOT_SERVICE
GND

External isolated fused 5V; separate peripheral 3V3; module 3V3 outputs never joined
Direct-solder factory male pins; no sockets; no carrier SMT components

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Title: MC1 Nano – ready-module carrier / engineering prototype

Size: A3	Date: 2026-10-17	**Rev: MC1-Nano**
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